

What is space?

Space holds many secrets. It contains places where human beings can be stretched into spaghetti shapes, or boiled, or frozen solid: that's why astronauts wear protective clothing in space. Welcome to a mysterious – and endlessly fascinating – world.

What is space?

When people think of space, they think of:



Weightlessness

– everything floats as if there's no gravity.



Nothingness

– vast areas of space are completely empty.



Stars – every star is a burning ball of gas. Our Sun is a star.



Astronauts – people who explore the world beyond our Earth.



Rockets and satellites – are what scientists use to explore space.



Silence – there is no air in space, so there is absolutely no sound.

A nebula is a cloud of dust and gas in space. This is the Helix nebula, about 700 light years away, seen from NASA's *Spitzer* Space telescope.



Is that space?

On a cloudless night, you can see thousands of stars. Space is the name we give to the huge empty areas in between the atmospheres of stars and planets. Apart from the odd rock, space is sprinkled only with dust and gas.

Too big to imagine

Astronomers measure distance in space in light years. One light year is the distance light travels in one year: that's 10 million million km (6 million million miles).